Reversibility

Ver. 20-10-2024

Reversibility is a central concept in thermodynamics. The term "*Reversible*" stems from the idea that some processes have the ability to reverse, i.e., to revert to the original state without any significant additional force applied.

A common example given to illustrate a reversible process is a ball on a strictly horizontal table. The ball may roll in one direction and may just as well roll back to where it came from. Here as always in thermodynamics, friction is disregarded, and the process is reversible. It is also infinitely slow, but that is not a worry in thermodynamics. Thermodynamics is an abstraction used for analysing the energy dynamics.

The corresponding irreversible process should then be a ball on a tilted table. This ball will spontaneously roll in one direction – downwards. We can make it roll back up by applying a force and do some work ion it, but it will not happen spontaneously.

An important feature of irreversible processes is that they generate entropy. We can actually define reversible processes as processes that don't generate entropy and irreversible processes as processes that generate entropy. This makes the claimed irreversible case with the ball rolling down a poor example, since it doesn't generate entropy. It merely converts potential energy into kinetic energy, and none of these two mechanical forms of energy represents any entropy (no spreading of matter or energy).

A better example of an irreversible process is the expansion of a gas into a larger volume. This happens spontaneously if a valve is opened, and we know for sure that the gas will not contract and flow back into the original volume without our significant interference with an amount of compression work. The gas in the large volume is more scattered and contains more entropy than in the first place before expansion.

Yet another example is a closed system with boiling water in equilibrium with water vapour. If heat is added to the system, a corresponding amount of liquid water will evaporate. Likewise, if we remove heat, some vapour will condense and add to the liquid pool. The process of evaporation and condensation is reversible itself, but the process of heat transfer is not necessarily. If transfer of heat happens without a temperature difference, then it may equally easily happen whether the direction is into or out of the system. It is reversible. The heat flow carries entropy, but it is not generated only transferred. A process can transfer entropy and still be reversible as long as no new entropy is generated. Irreversibility causes generation of entropy, not just transfer.

This is important and we can now come back to the boiling water. The driving force for heat transfer is a temperature difference. Reversible transfer of heat happens without a temperature difference and thus without a driving force and it is infinitively slow. Again, not problem in thermodynamics, but in real systems, we cannot wait forever, and we make use of a temperature difference. The hotplate under the system is at a higher temperature. This makes thing happen, but the process is then irreversible. The revers process would require that the heat liberated from condensation should be transferred back into the hotplate. This will not happen.

In general, one can state that when a net driving force is in action, the process driven is irreversible. Real processes are therefore always irreversible although they can be "almost reversible". The degree of irreversibility can be quantified by the production of entropy. Common driving forces are

- Temperature differences, driving heat transfer
- Pressure differences, driving movement
- Electrical potential differences, driving electrical current
- Chemical potential differences, driving chemical processes
- Concentration differences, driving diffusion